

Machine Learning for Natural Language Understanding

Exercise WS 2024/2025

Mandatory Exercise

Due: 27.02.2022

Multiple Choice**10 Points**

Are the following statements true or false?

	Statement	True	False
1	The sigmoid-function $h_{\theta}(x)$ is smooth and symmetric at $x = 0$.	X	
2	BERT is a language model based on RNNs		X
3	A Multilayer Perceptron can only represent simple linear functions		X
4	Every continuous function can be approximated arbitrarily closely by a multi-layer Artificial Neural Network.	X	
5	GPT-3 is known for its language generation abilities	X	
6	RNNs capture long-term dependencies		X
7	Hyperparameters should be tuned on the validation set	X	
8	An overfitted model performs well on unknown data		X
9	The classification of unbalanced data is measured best with accuracy		X
10	The harmonic mean of precision and recall is called F-measure	X	

Aspects of Machine Learning Models**10 Points**

1) What is the term for the process of converting real-world data into representations that can be used, for example, as input for neural networks? Provide an example! (2 Points)

The term is *feature extraction*

Examples are:

1. TF-IDF
2. W2V

2) Explain what the training, validation and test data sets are used for in machine learning. (3 Points)

- Training set for training, model is fit on
- Validation set used for tuning parameters, not for training
- Test set: provide an unbiased evaluation of final model

3) What is the difference between *supervised*, *unsupervised* and *reinforcement* learning? (3 Points)

- Supervised learning is a machine learning algorithm of inferring a function from labeled training data. The training data consists of a set of training examples.
- Unsupervised learning is also a type of machine learning algorithm used to find patterns on the set of data given. In this, we don't have any dependent variable or label to predict.
- Reinforcement learning: learns to react to an environment based on a reward function

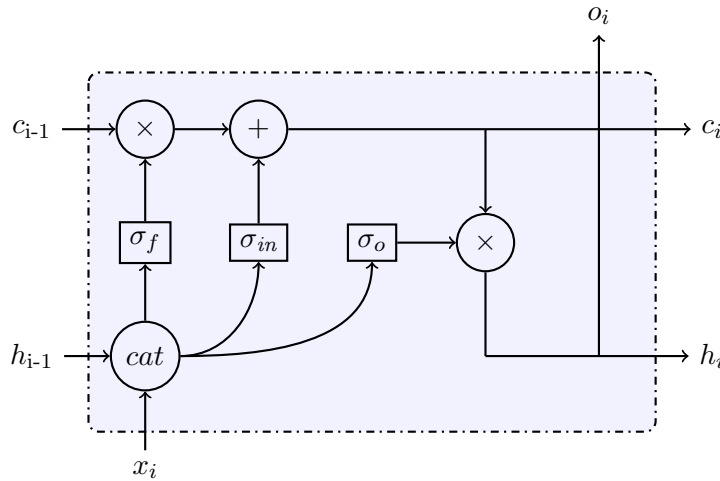
4) Explain the difference between regression and classification? (2 Points)

1. Regression: Works on continuous values; approximates a real-valued function
2. Classification: Works on discrete values

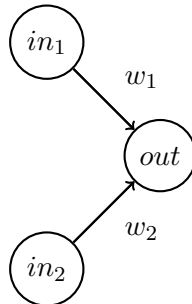
Neural Networks I

10 Points

1) Consider this simplified LSTM cell:



Assume all three gates (σ_x) to be simple feed-forward networks with two inputs and one output:



Assume the Bias to be 0. *Cat* stands for the concatenation operation in the following order: $(cat(h_{i-1}, x_i))$. **Example:** $x_1 = 5, x_2 = 3 : cat(x_1, x_2) \rightarrow [1, 5]$.

$\times$ and $+$ are multiplication and addition. Assume the following activation function for all σ :

$$f(x) = \begin{cases} 2, & \text{if } x \geq 2 \\ 1, & \text{if } x < 2 \end{cases}$$

Given the following weights and inputs, calculate the values for o_i , c_i and h_i (6 points)!

INPUTS:

$$h_{i-1} = 2$$

$$c_{i-1} = 1$$

$$x_i = 1.5$$

WEIGHTS:

$$w_{f1} = 0, w_{f2} = 0.6$$

$$w_{in1} = -0.5, w_{in2} = 0.5$$

$$w_{o1} = 0.5, w_{o2} = 0.25$$

- SEE EXERCISE SHEET - ONLY THE VALUES WERE DIFFERENT

2) Explain how the LSTM architecture prevents Gradients from vanishing. (2 Points)

- By introducing the cell state, which provides a backpropagation path without nonlinearities
- The cell state is updated through a gating mechanism, meaning that new information can be added and old information deleted

3) Explain the main disadvantage of LSTMs when compared to the Transformer architecture. (In terms of scalability) (2 Points)

- An LSTM (and RNNs in general) processes sentences word by word (token by token). All intermediate steps must be calculated before the next step can be calculated
- A Transformer processes a sentence (or a sequence in general) at once. Therefore, it becomes highly parallelizable.

Neural Networks II**10 Points**

1) Name 5 commonly used activation functions! Indicate the respective value ranges. *(5 Points)*

- tanh $-1, 1$
- ReLU $0, \infty$
- softmax $0, 1$
- sigmoid $-e, \infty$
- step $0, 1$

2) What is the *Vanishing Gradient Problem*? *(2 Points)*

- Vanishing gradients occur due to the application of the chain rule in backpropagation. By multiplying the derivatives of many activation functions, the gradients can approximate zero and thus will carry next to zero information.

3) Why is it hard to find a fitting learning rate for machine learning model training? What are the possible outcomes of a badly chosen learning rate and how can this problem be tackled? *(3 Points)*

- Too high: 'Overshooting' the solution. With a large learning rate, it might be impossible for the algorithm to reach an optimum
- Too low: 'Getting caught' in a local optimum. With a small step size, it might become impossible for the algorithm to get out of a local optimum, thus making it impossible to find the best solution
- Adaptive learning rate: By switching from a fixed to an adaptive learning rate, the step size can be changed depending on the current learning progress

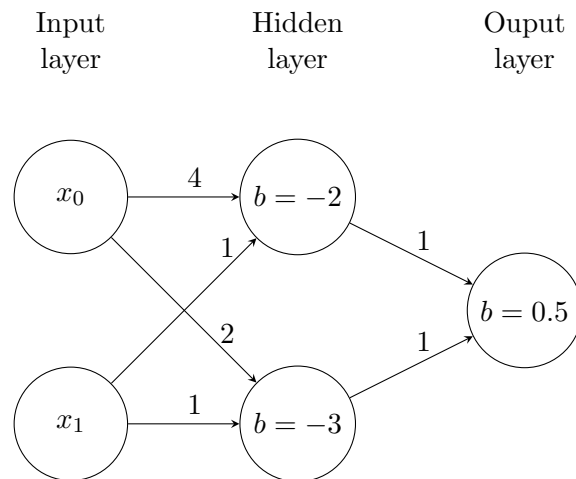
Neural Networks III**10 Points**

1) The following table contains predicted values from a simplified linear model ($Y_i = \text{Beta}_0 + x_i$) and their true (i.e. expected) counterpart. Show the calculation of the MSE for this model! Which value should Beta_0 have to minimize the MSE? (5 Points)

Predicted	Expected
0	1
1	2
3	4
4	5

- $\frac{(0-1)^2 + (1-2)^2 + (3-4)^2 + (4-5)^2}{4} = 1$
- Beta_0 needs to be 1

2) Take a look at the following Neural Network:



Calculate the network predictions and compare them with the expected classes. Assume *act* as activation function. (Apply the bias before the activation) (5 Points)

x_0	x_1	Class
1	1	1
2	1	1
-3	2	-1

$$\text{act}(x) := \begin{cases} +1, & \text{if } x > 0, \\ -1, & \text{if } x \leq 0. \end{cases}$$

SEE EXERCISE SHEETS

Language Models**10 Points**

- 1) Name and describe the two most common pre-training tasks for Language Models *(2 Points)*
 - Masked language modelling: The task is to predict a masked word given its context
 - Next word prediction: The task is to predict the next word in a give sequence
- 2) What are positional embeddings and why are they used in the context of Transformer models? *(2 Points)*
 - Positional encodings are vector representations of positions. They are designed such that the vector for position 1 is closer to the vector for position 2 than the one for 3.
 - They are used because the transformer architecture in its own has no notion of order (all words are processed in parallel)
- 3) Name at least four downstream tasks at token or text level and briefly explain them. *(4 Points)*
 - Named entity recognition: Token level task that marks tokens describing a real-world entity
 - Question answering: Token or Text-level task that answers to a given question
 - Text Classification: Text-level task that generates a summary of a given text
 - Natural Language Inference: Determines whether a hypothesis is entailed in a premise
- 4) Explain the differences between static and contextualized word embeddings. *(2 Points)*
 - Static Word Embeddings: One token has one fixed representation. They are independent on the current context and therefore unable to disambiguate different meanings of a word
 - Contextualized Word Embeddings: Depending on the current context, the vector representation of a token may change, allowing to represent the different meanings of a word given its context.